Problem 1. Prove that n^2 + n is even for every integer n.

For n = 1 we get 1 + 1 = 2. For n = 2 we get 4 + 2 = 6. For n = 3 we get 9 + 3 = 12.

All of these are even, so n^2 + n is even for every integer n.